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BIG DATA ANALYTICS LAB

Exp-4. Working with Hive on the following

ii) Writing User Defined Functions in Hive

Date: 20-02-2025

Step 1: Write the Hive UDF in Python

Create a Python script (`strip_udf.py`) that removes characters from both ends of a string.

hadoop@ubuntu22:nano /home/hadoop/strip_udf.py

```
#!/usr/bin/env python3
import sys

# Read input line by line
for line in sys.stdin:
    # Remove leading/trailing whitespace and strip special characters
    print(line.strip().strip('.,!'))
```

Step 2: Make the Python Script Executable

hadoop@ubuntu22:chmod +x /home/hadoop/strip_udf.py

```
hadoop@ubuntu22:~$ chmod +x /home/hadoop/strip_udf.py
hadoop@ubuntu22:~$
```

Step 3: Upload the File to HDFS (Optional)

If needed, upload it to HDFS:

hadoop@ubuntu22:hdfs dfs -put /home/hadoop/strip_udf.py /user/hadoop/

```
hadoop@ubuntu22:~$ hdfs dfs -put /home/hadoop/strip_udf.py /user/hadoop/
put: `/user/hadoop/strip_udf.py': File exists
```

Step 4: Start the Hive Shell

hadoop@ubuntu22:~$ cd $HIVE_HOME

hadoop@ubuntu22:~/apache-hive-3.1.3-bin$

hadoop@ubuntu22:~/apache-hive-3.1.3-bin$ hive

```
hadoop@ubuntu22:~$ cd $HIVE_HOME
hadoop@ubuntu22:~/apache-hive-3.1.3-bin$ hive
```

Step 5: Add the Python UDF File in Hive

Inside the **Hive shell**, run:

ADD FILE /home/hadoop/strip_udf.py;

To verify the file is added, run:

```
hive> ADD FILE /home/hadoop/strip_udf.py;
Added resources: [/home/hadoop/strip_udf.py]
hive>
```

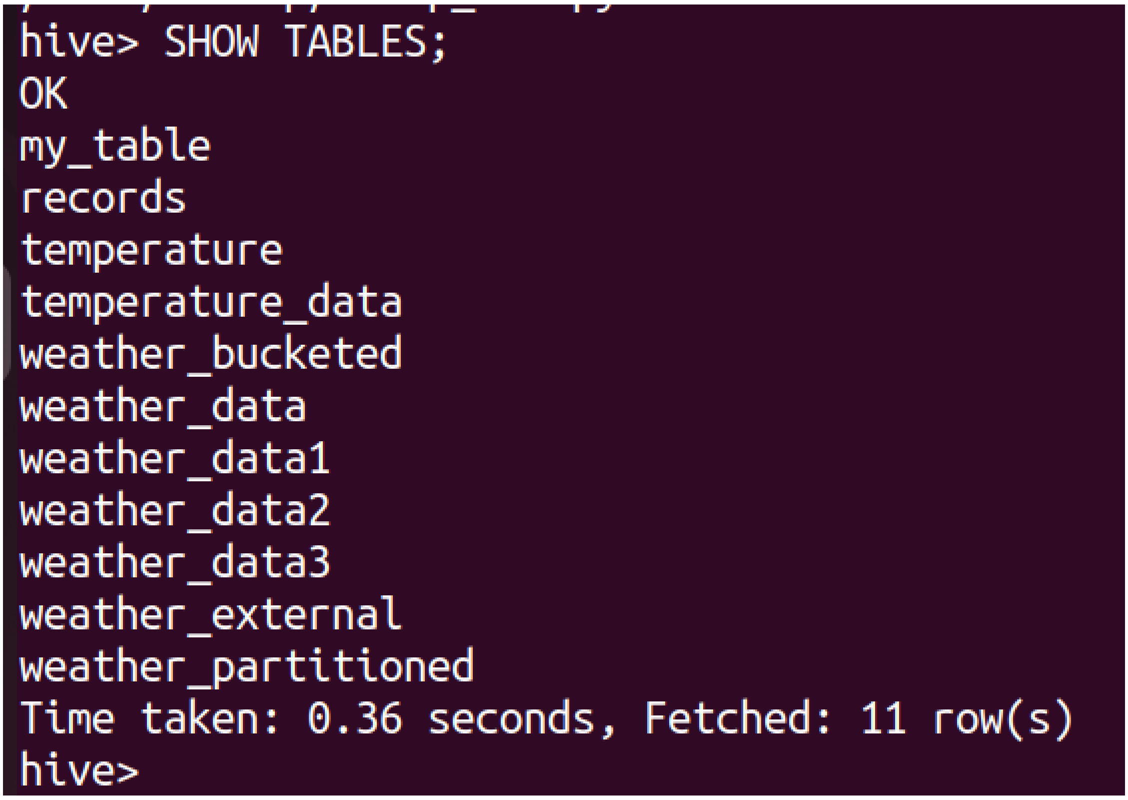
hive> LIST FILE;

```
hive> LIST FILE;
/home/hadoop/strip_udf.py
hive>
```

Step 6: Ensure Your Table Exists

Check if your table exists:

hive>SHOW TABLES;



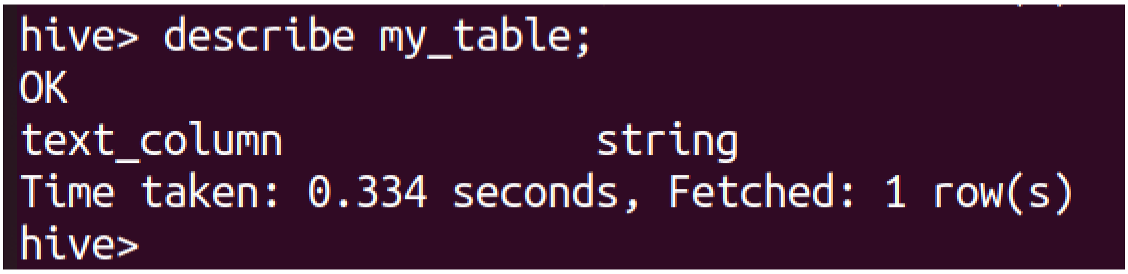
```
hive> SHOW TABLES;
OK
my_table
records
temperature
temperature_data
weather_bucketed
weather_data
weather_data1
weather_data2
weather_data3
weather_external
weather_partitioned
Time taken: 0.36 seconds, Fetched: 11 row(s)
hive>
```

If my_table is missing, create it:

hive> CREATE TABLE my_table (text_column STRING);

hive> describe my_table;
OK
text_column   string
Time taken: 0.334 seconds, Fetched: 1 row(s)

hive>



```
hive> describe my_table;
OK
text_column                 string
Time taken: 0.334 seconds, Fetched: 1 row(s)
hive>
```

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/sample.txt' INTO TABLE my_table1;
```

sample.txt:

```
 Hello World!!!
,Data Processing.,
  Big Data!!
Python,Hadoop!!
```

```
hive> select * from my_table1;
OK
 Hello World!!!
,Data Processing.,
  Big Data!!
Python,Hadoop!!

Time taken: 0.837 seconds, Fetched: 5 row(s)
```

```
hive> CREATE TABLE my_table1 (text_column STRING);
OK
Time taken: 0.679 seconds
hive> LOAD DATA LOCAL INPATH '/home/hadoop/sample.txt' INTO TABLE my_table1;
Loading data to table default.my_table1
OK
Time taken: 0.644 seconds
hive> select * from my_table1;
OK
 Hello World!!!
,Data Processing.,
  Big Data!!
Python,Hadoop!!

Time taken: 0.837 seconds, Fetched: 5 row(s)
hive>
```

Step 7: Run the Hive Query with the Python UDF

Execute:

```
hive> SELECT TRANSFORM (text_column)
    > USING 'python3 strip_udf.py'
    > AS (cleaned_text)
    > FROM my_table1;

OK
Hello World
Data Processing
Big Data
Python,Hadoop

Time taken: 1.743 seconds, Fetched: 5 row(s)
```

```
hive> SELECT TRANSFORM (text_column)
    > USING 'python3 strip_udf.py'
    > AS (cleaned_text)
    > FROM my_table1;
Query ID = hadoop_20250220121616_a5ccb978-7e48-44d1-8dca-725cc9456262
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks is set to 0 since there's no reduce operator
Job running in-process (local Hadoop)
2025-02-20 12:16:17,966 Stage-1 map = 100%,  reduce = 0%
Ended Job = job_local1748679448_0002
MapReduce Jobs Launched:
Stage-Stage-1:  HDFS Read: 198 HDFS Write: 66 SUCCESS
Total MapReduce CPU Time Spent: 0 msec
OK
Hello World
Data Processing
Big Data
Python,Hadoop

Time taken: 1.743 seconds, Fetched: 5 row(s)
hive> hadoop@ubuntu22:~/apache-hive-3.1.3-bin$
```